Qualitative Analysis with TAGUETTE

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About me!

- Day job: Data librarian at NYU Data Services
- Subject specialist for computer science and data science
- Open everything ~
- she/her/hers or they/them/theirs

- Taguette is my nights and weekends project with my husband Rémi 🖤 who is the main developer

Access to CAQDAS 🌵

- Number of free and open source CAQDAS has been and is LOW (< 20)
- Perpetual access to qualitative software are very expensive for individual users
  - $689 MaxQDA, $1,019 Nvivo, $150-340 Quirkos
- Some CAQDAS have gone to a subscription model for their Web and desktop versions:
  - Atlas.TI $20-357/mo, Quirkos $16/mo, Dedoose $13-18/mo

Comparison of CAQDAS software [edit]

Application	Type	License	Source
QualCoder	Client	Free - MIT License	Open
ATLAS.ti	Client	Proprietary	Closed
Quirkos	Client / Web-based	Proprietary	Closed
Dovetail	Web-based	Proprietary	Closed
NVivo	Client	Proprietary	Closed
CLAN	Client	Free – GPL	Open
MAXQDA	Client	Proprietary	Closed

en.wikipedia.org/wiki/Computer-assisted_qualitative_data_analysis_software

Why free and open source software matters

Equity

Researchers without access to funding deserve CAQDAS

Avoiding lock-in

You can take the software with you wherever

Standardization

Open projects support interoperability

Prioritizes community

FLOSS software comes from community needs

TAGUETTE - web-based & collaborative

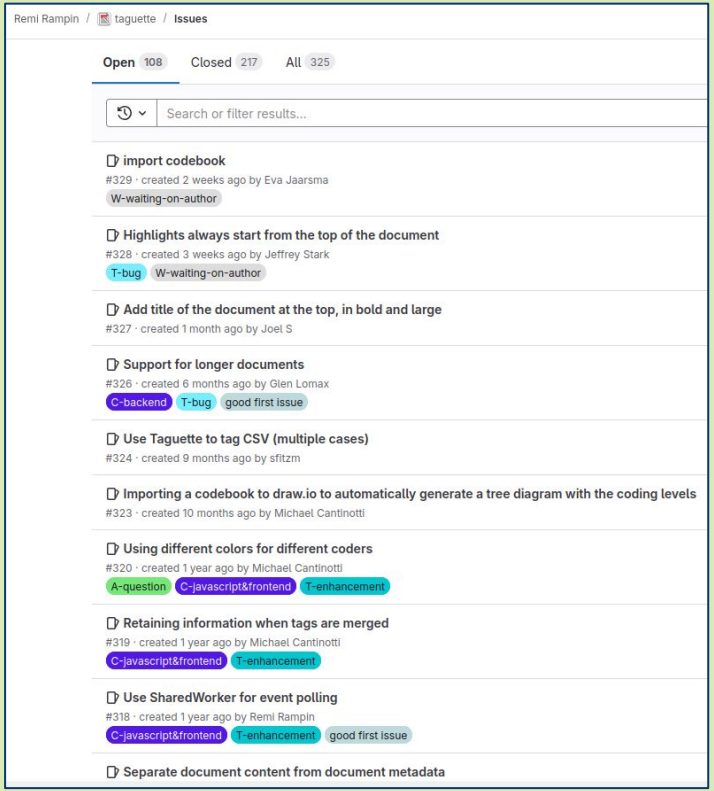
- Like “baguette” but with a “t” -- a play on the phrase “tag it!”
- Purely text tagging, desktop or server
- Development on GitLab: gitlab.com/remram44/taguette
- Remi runs a hosted version of Taguette for users: app.taguette.org

TAGUETTE

FEEDING YOUR QUALITATIVE NEEDS

Technical approach to building TAGUETTE

- Built using Python 3, SQLite, & Calibre (FLOSS ebook manager)
  - Calibre is used to convert uploaded materials to HTML
  - wv used to convert old .doc files
- All data stored in SQLite database by default, but can be changed to a database of your choice if you want



TAGUETTE Features

Available on all OS

It works the same on all operating systems!

Real-time collaboration

On a server, you can collaborate in real-time

Easy to learn & use

User interface is simplified to help folks navigate

Self-hostable

Anyone can host an instance of Taguette, useful for institutions or those with high-security needs

Data remains local

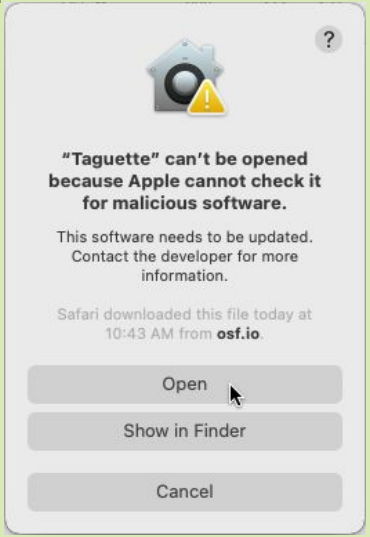
If using on a desktop, data remains local to computer

Multilingual

Thanks to volunteers, multiple languages available

Installing TAGUETTE on your local computer

- Install instructions at: taguette.org/install.html
  - There are some special steps, so be sure to read through the guide!
- Installers for macOS & Windows also on GitLab: gitlab.com/remram44/taguette/-/releases/v1.4.1
- "Taguette" entry should appear in your list of applications:
  - Start Menu on Windows
  - Launchpad/Spotlight on macOS



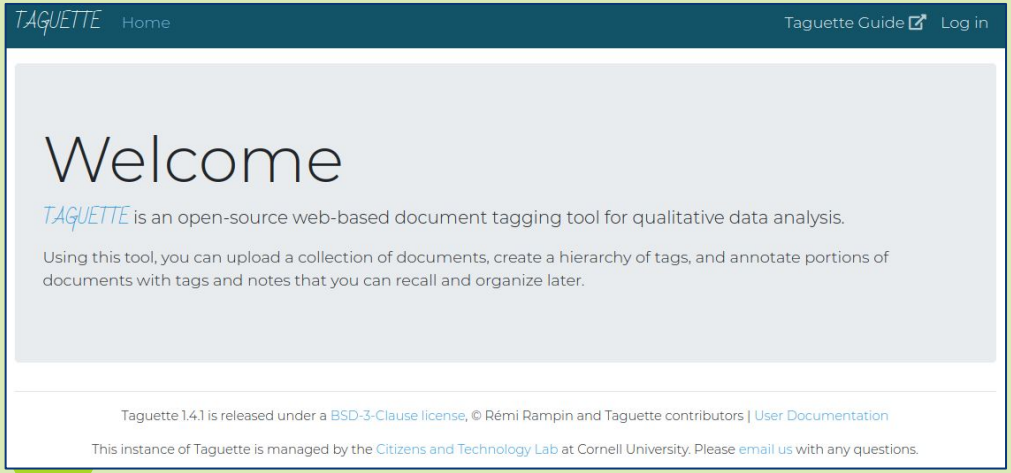
Install TAGUETTE on your own server

You can install Taguette on a server of your own choice, or you should ask if an IT unit in your organization will, info in our self-hosting guide!

This is particularly useful for secure and collaborative QDA

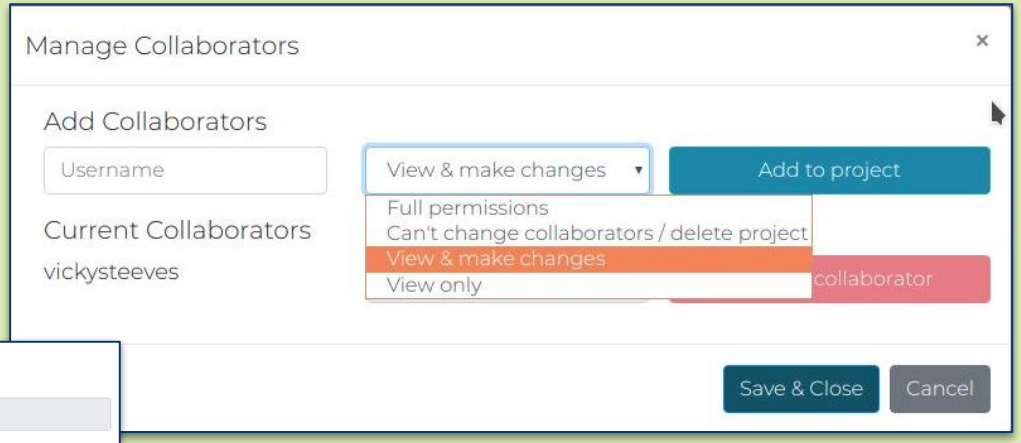
Some examples:

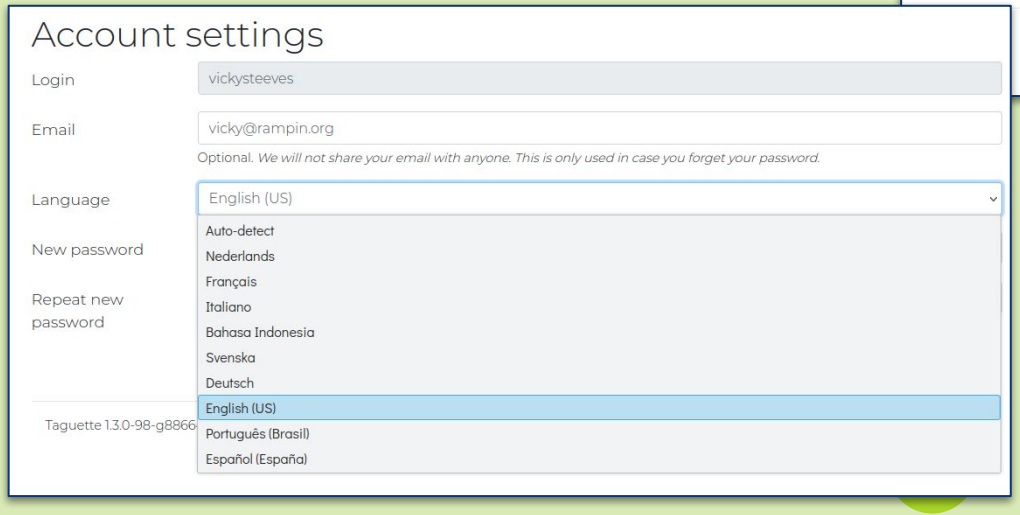
- taguette.cs.ksu.edu
- taguette.citizensandtech.org
- taguette.lehman.edu
- taguette.monmouth.edu
- cpec.tag.lse.ac.uk
- taguette.cs.uwaterloo.ca



Features only on the server-version of TAGUETTE

Adding collaborators (right): since the server is multi-user, you can add others to specific projects with varying levels of permissions

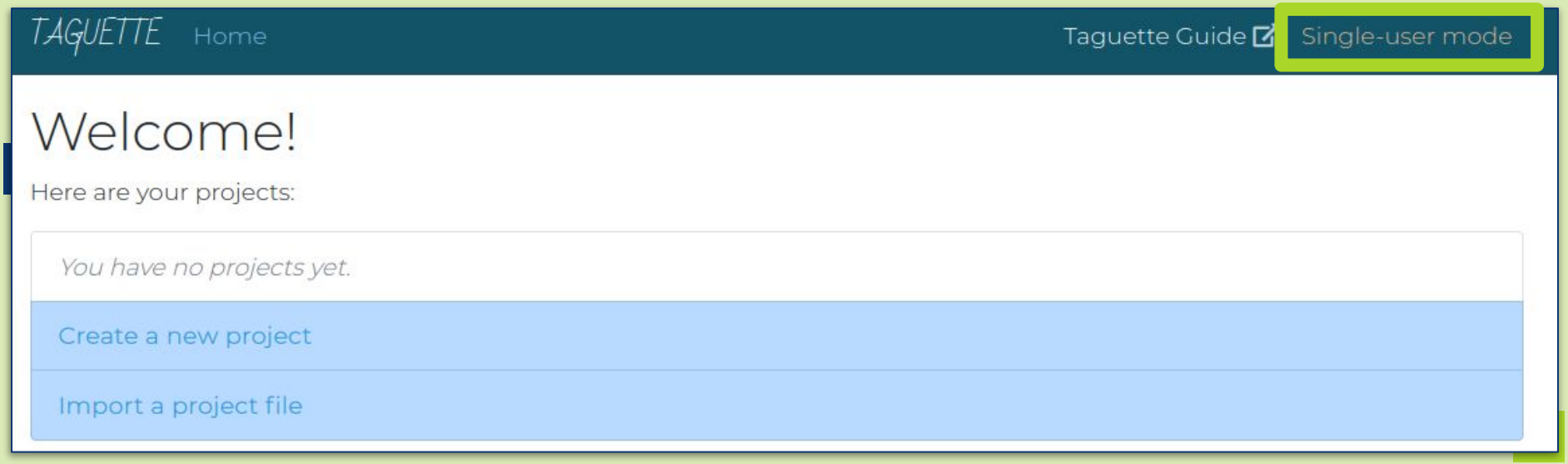




Button for switching languages (left): generous volunteers have translated Taguette's interface into multiple languages. Contribute & view progress at transifex.com/remram44/taguette

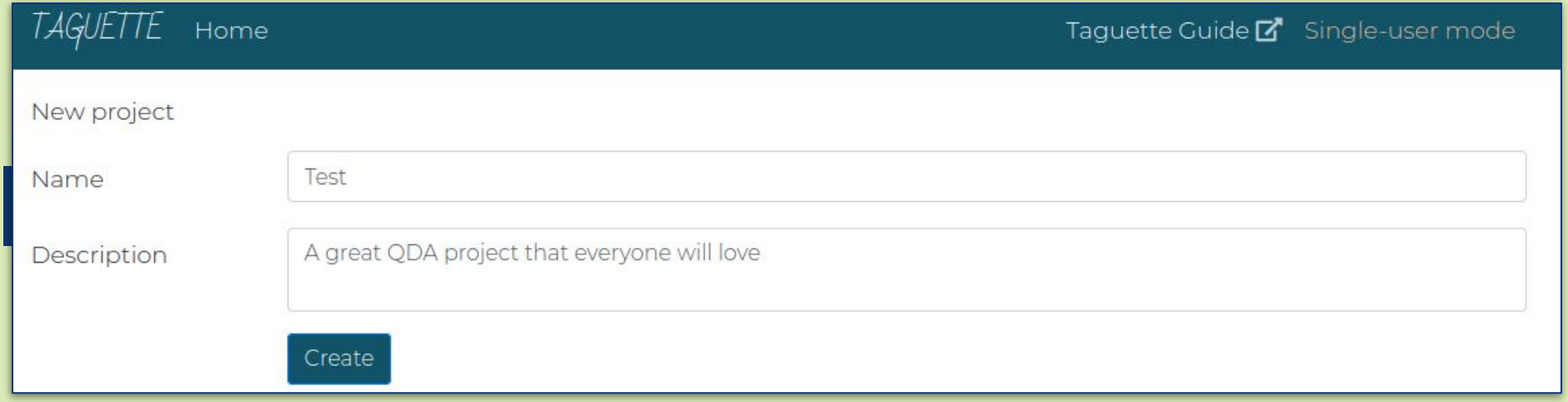
Starting Up TAGUETTE

When you start it on your desktop, you'll see "single-user mode" in the corner:



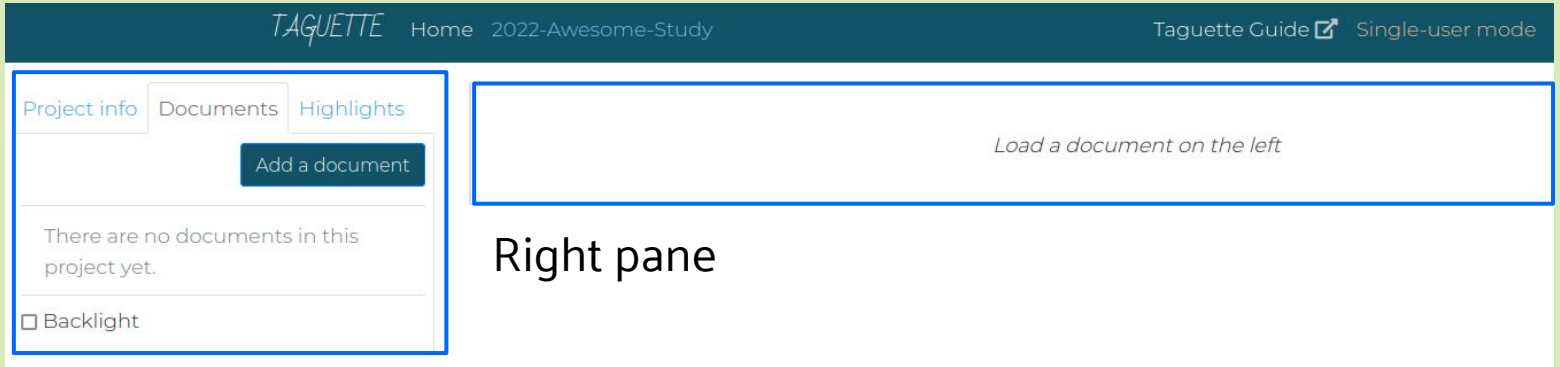
Creating a new project in TAGUETTE

- After you click **Create a project**, you'll be prompted to enter a **Title** and **Description**
  - This can be changed later on



Starting to work in TAGUETTE

- After clicking Create, you'll be taken the Project View, which has a left and a right pane:
  - Left pane contains the information about your project information ('Project Info'), uploaded materials ('Documents'), and tags ('Highlights') as tabs.
  - Right pane will render documents and highlights and where you export those
    - It will also be the area where you'll do the highlighting and tagging of docs

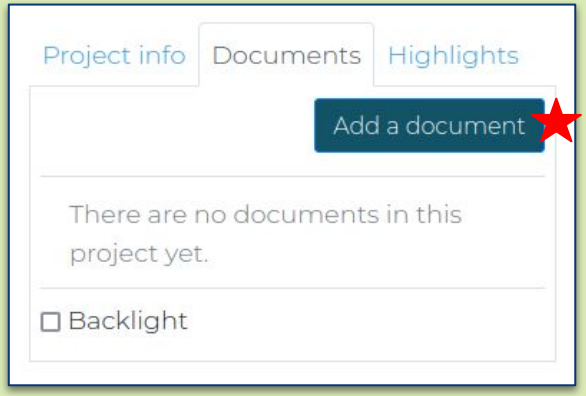


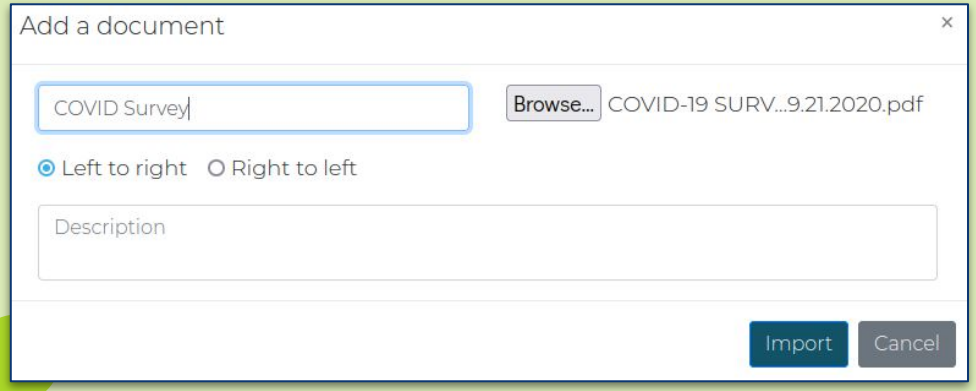
Right pane

Left pane

Adding a document in TAGUETTE

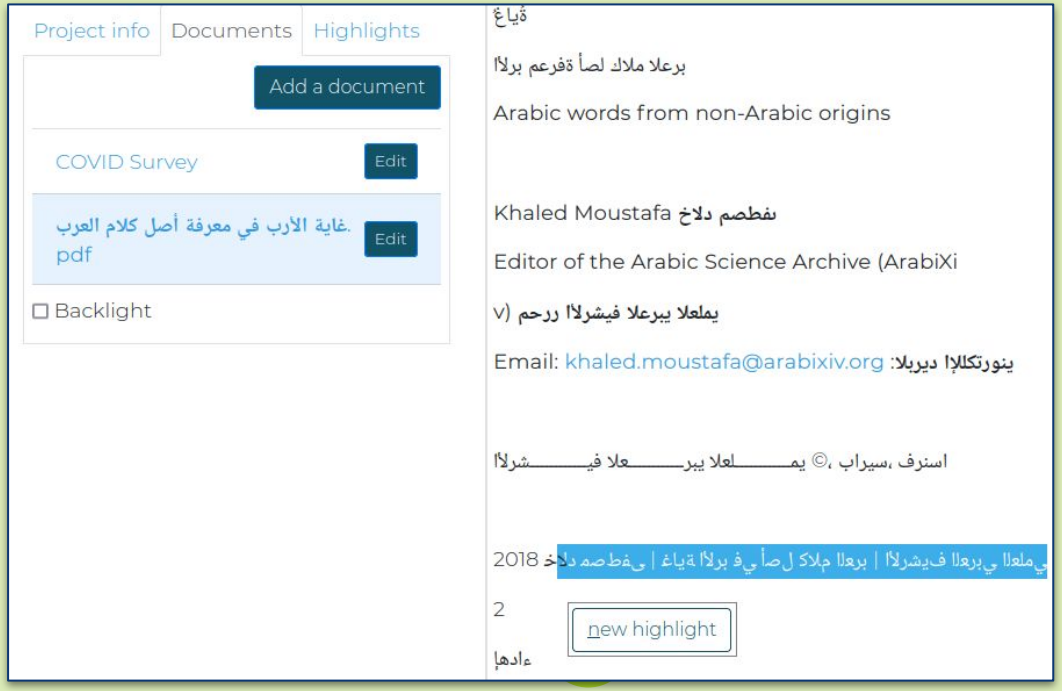
- In the left pane, click on the Documents tab. You should see a button that says **Add a document**. Click that and pick just one document from your computer.
  - Taguette supports <u>everything Calibre supports</u>, plus .doc files
- Mark the document as left-to-right or right-to-left
- Provide a description of the document for yourself (optional)





Multilingual support for source materials

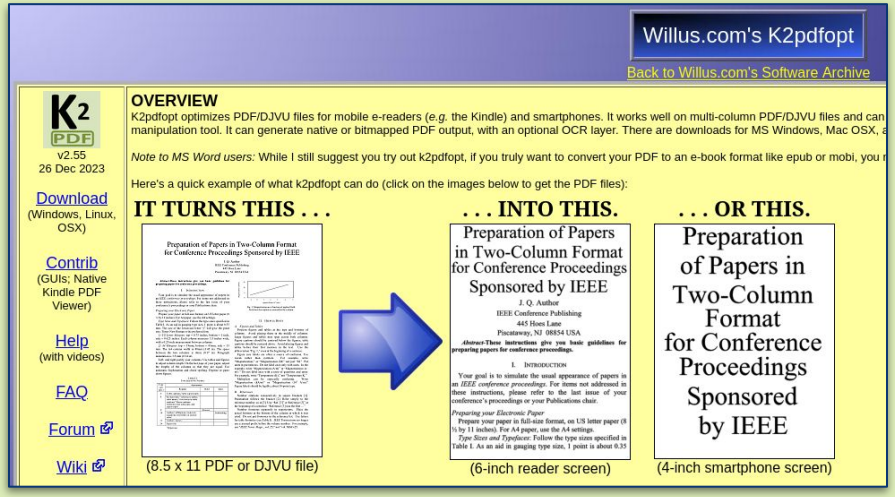
- Users can upload materials in any language they choose, including those right-to-left! Below is an example of Arabic text uploaded to Taguette:



Warning about document structure & import

There are some known issues with importing we are working on:

- 2 column documents often get text jumbled up between lines
  - Need to convert these to 1 col
- Non-OCR'd PDFs will be treated as images and imported as blank
  - Need to OCR these before importing

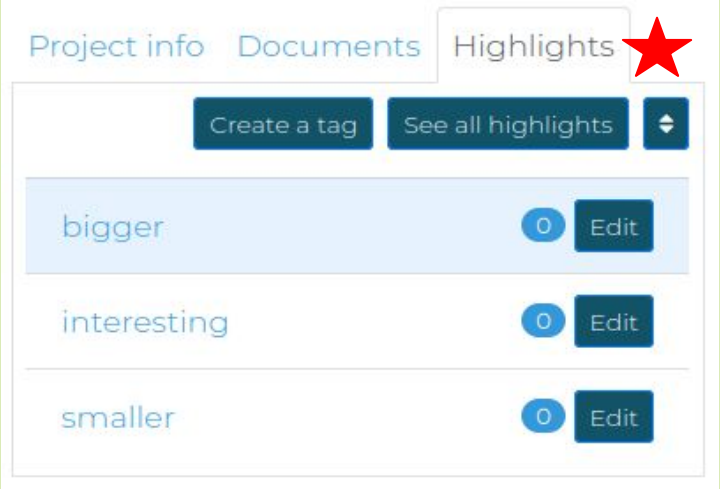


k2pdfopt works across OS

Viewing all tags in TAGUETTE

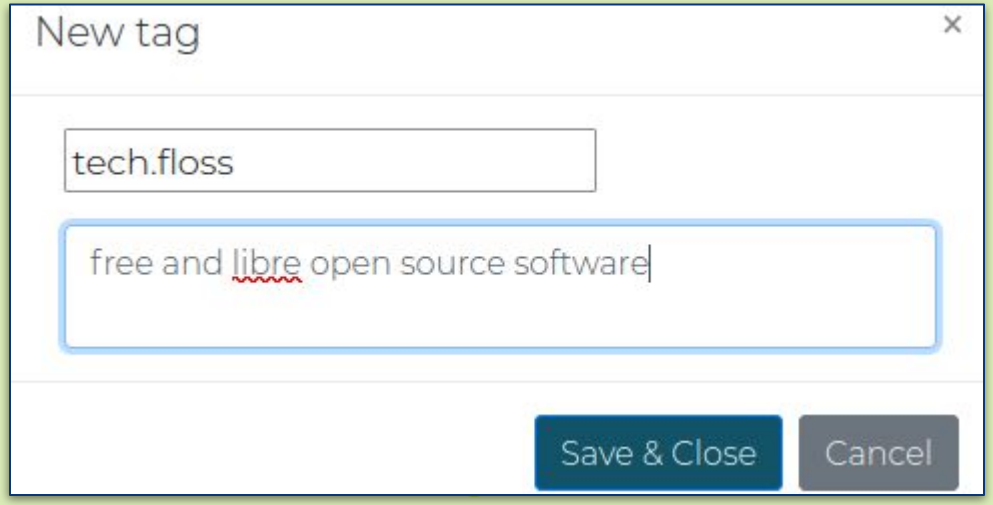
Taguette ships with one existing **tag** (or code): *interesting*.

To view all your existing tags, click the Highlights tab in the left pane.



Adding a code in TAGUETTE

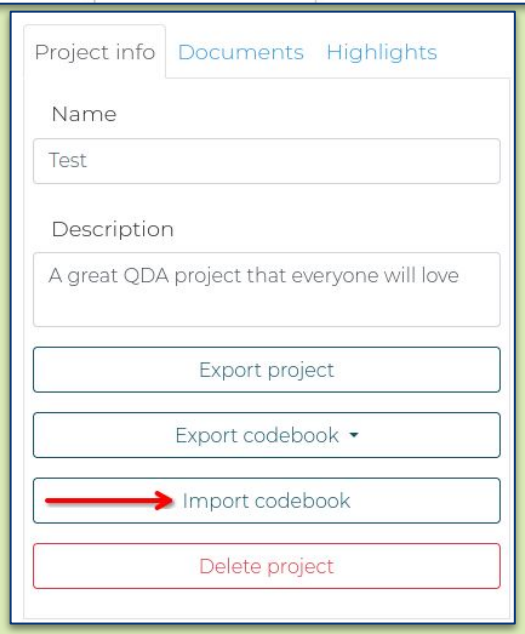
- To add your own tags to this project, click **Create a tag**
  - A modal will appear for you to enter the tag's **Name** and **Description**
- Be consistent and give an adequate amount of description -- it will help when you export a codebook later on!
- CTRL+F works if you want to find specific keyword(s) but you still code manually



Import a codebook

- If you have a codebook, you can import it into Taguette as a CSV
  - One column for the name of the tag, called: name, tag, path
  - One column: description
  - Any other columns are ignored
- Existing tags will not be deleted, new tags will be added alongside them
- **Project tab > Import codebook**

	A	B	C	D
1	tag	description		
2	interesting	something to note for later		
3	floss	free and libre open source software		
4	floss.tech	specific floss tools mentioned		
5	gender	when gender is explicitly discussed		
6	gender.woman	when womanhood is explicitly discussed		



Import a codebook

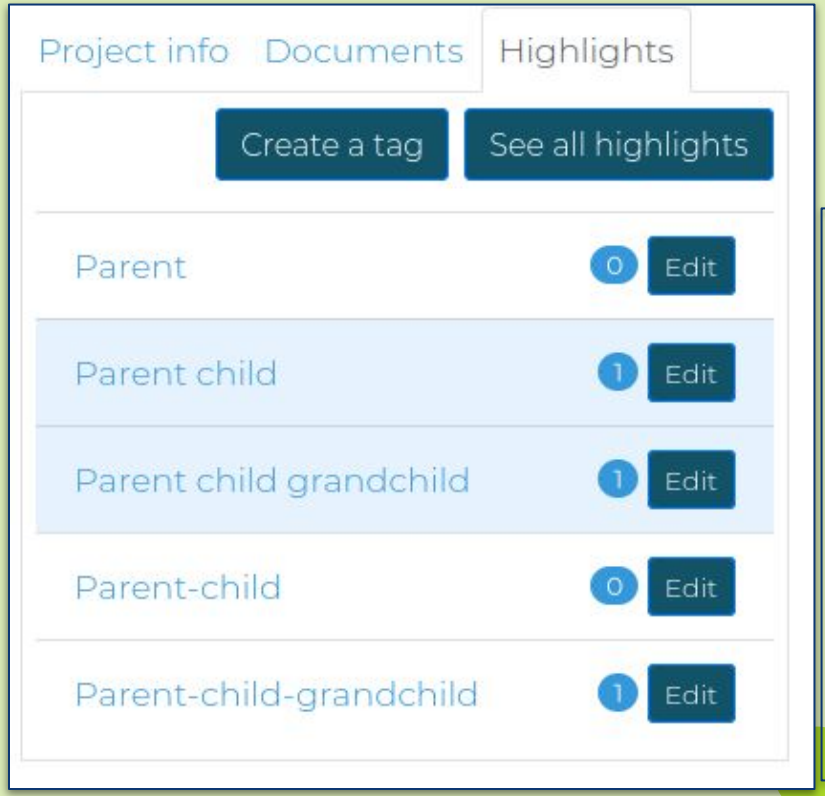
Use this form to import an existing codebook into your project. Existing tags will not be deleted, new tags will be added alongside them.

You are about to create the following tags:

☑ Import	floss	free and libre open source software
☑ Import	floss.tech	specific floss tools mentioned
☑ Import	gender	when gender is explicitly discussed
☑ Import	gender.woman	when womanhood is explicitly discussed
☑ Import	interesting	something to note for later

Create tags Go back

TAGUETTE DOES have hierarchical coding...it's just not visually a tree (yet)



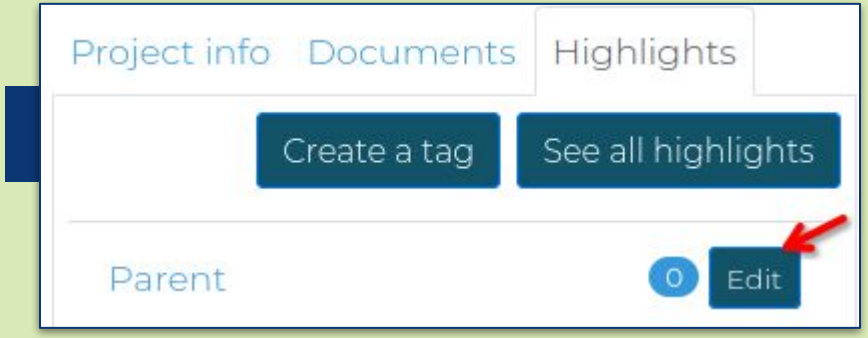
any space or punctuation makes a child tag -- and you need a separate Parent tag to view all it's children and grandchildren

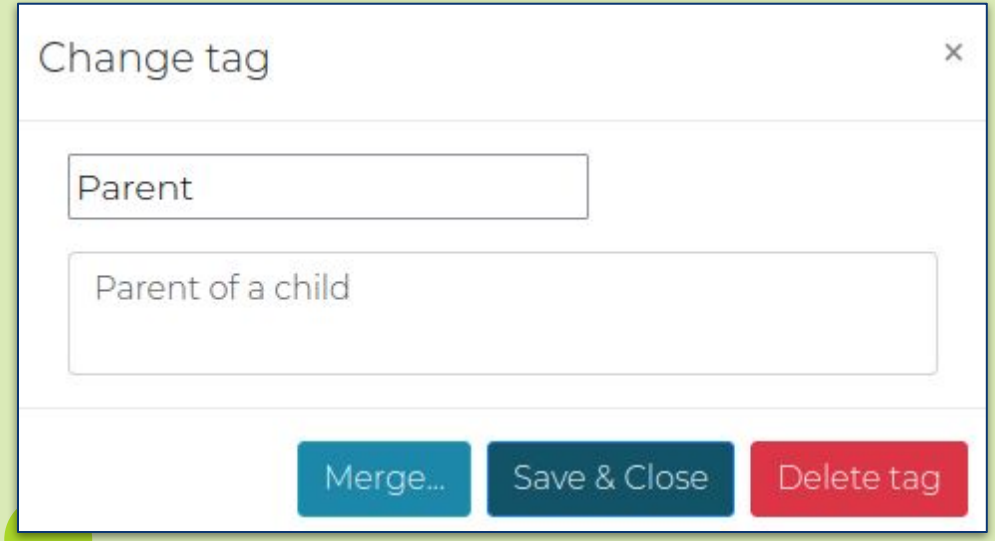
Parent

- Parent-child
  - Parent-child-grandchild
- Parent.child
  - Parent.child.grandchild
- Parent child
  - Parent child grandchild

Editing or deleting a code in TAGUETTE

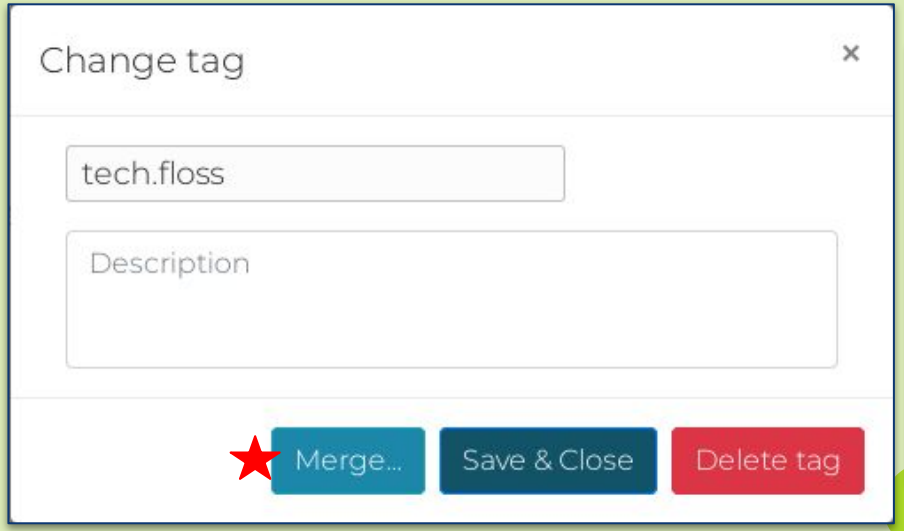
- Click the **edit** badge to the right of name of the tag in the list.
  - Modal will appear and you can edit the name or description
  - Click the grey Delete button to delete it altogether

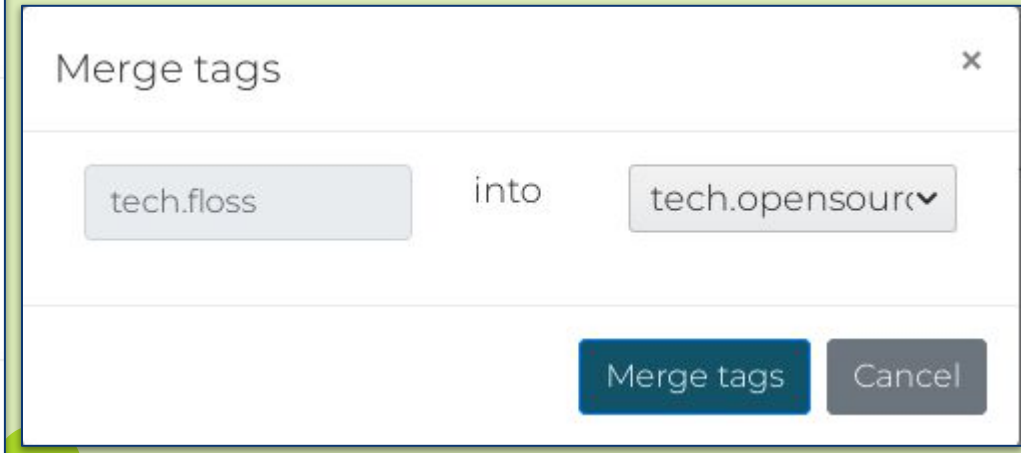




Merge tags together

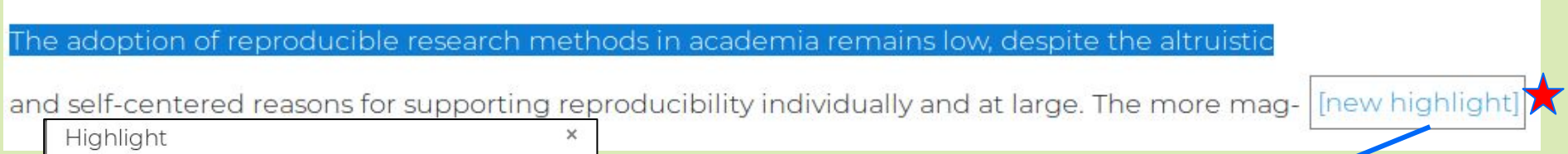
- Click the **edit** badge to the right of name of the tag in the list.
- Click light blue **Merge** button on the far left of the button options
- Select which tag you want to merge this one into from a dropdown list of all the tags in this project

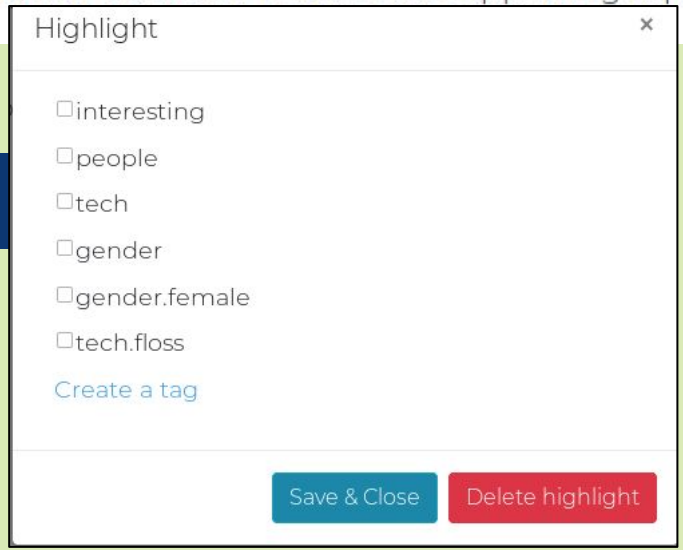




Highlighting documents in TAGUETTE

Select some text by left-clicking and dragging it over the text you'd like to highlight. Once you let go of your left-click, a pop-up that says new highlight will appear next to the highlighted text:

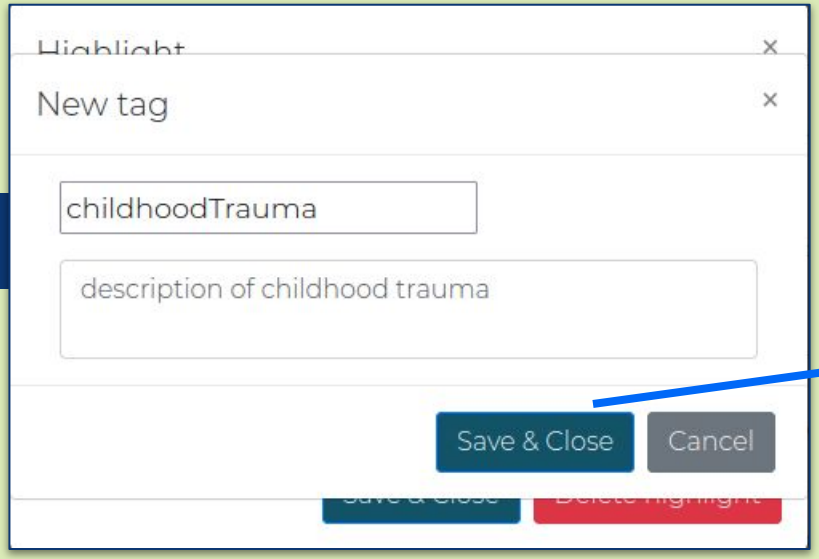


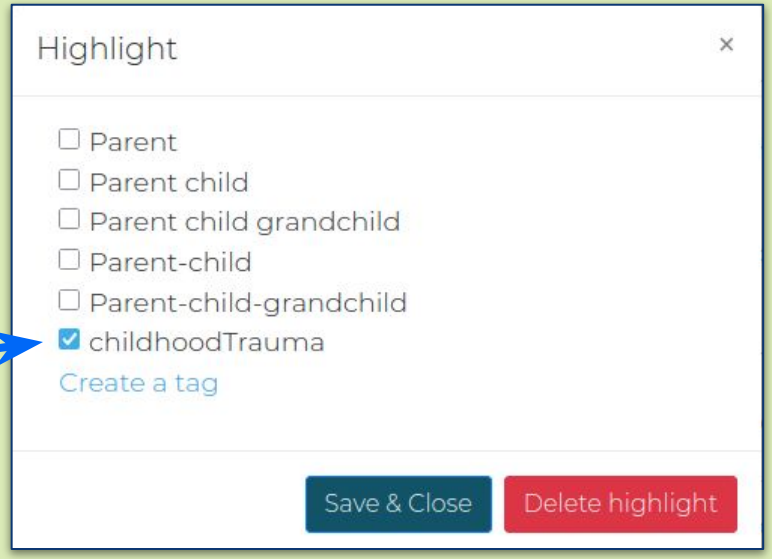


Click that pop-up and you can choose a tag to associate with this highlighted text. You can also create a new tag if you need to from this modal

Adding tags on the fly in TAGUETTE

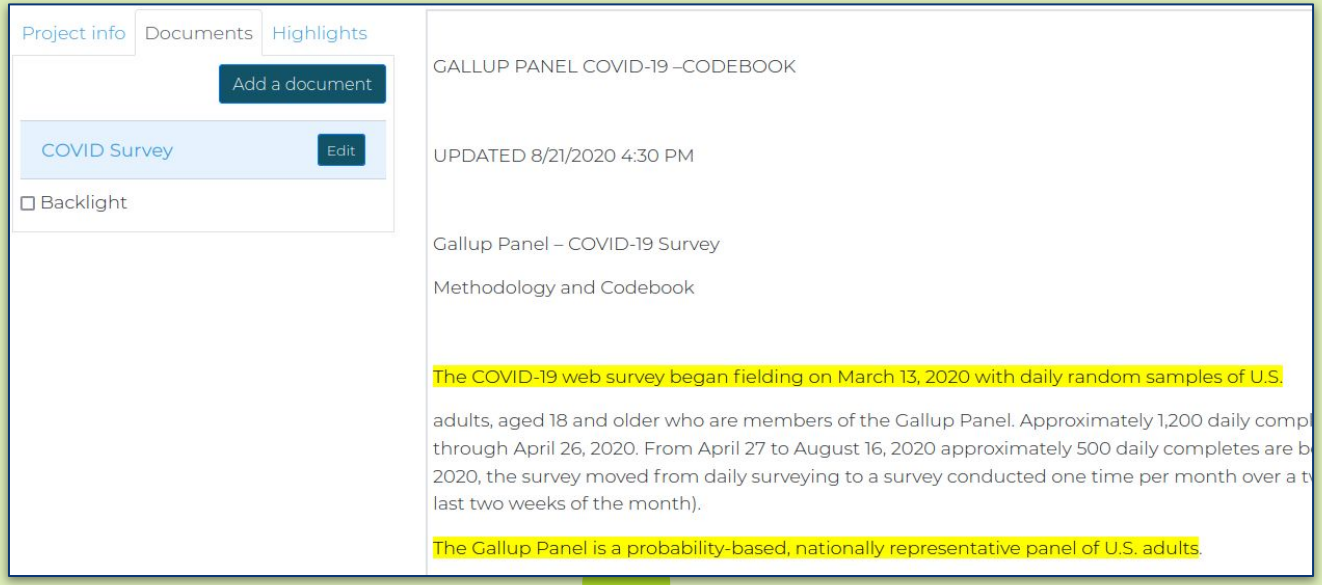
Clicking **Create a tag** will have another popup appear over your highlight popup where you can create a tag. Make sure to **Save & Close**. Then the modal disappears and your new tag will be auto-selected in the previous Highlight popup





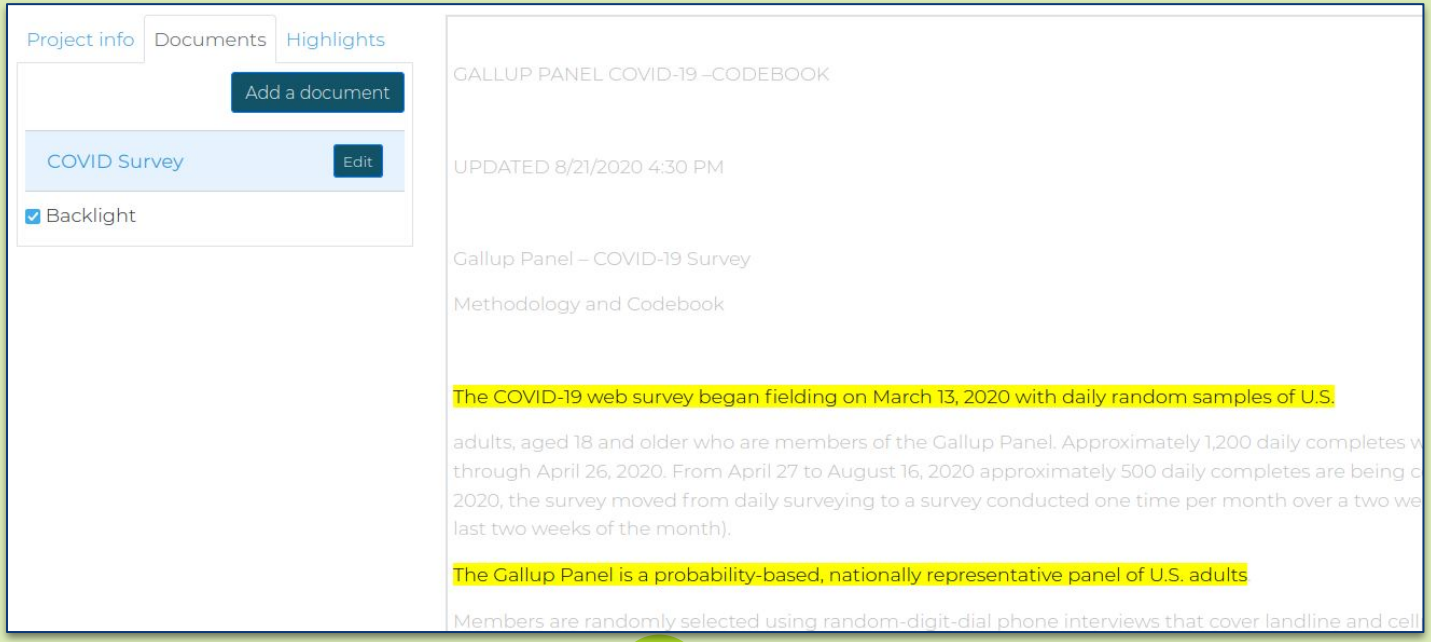
Seeing our work in the doc in TAGUETTE

Click the Documents tab in the left pane.
Click on the document you wish to view, and if you've highlighted it, you should see those highlights in yellow.



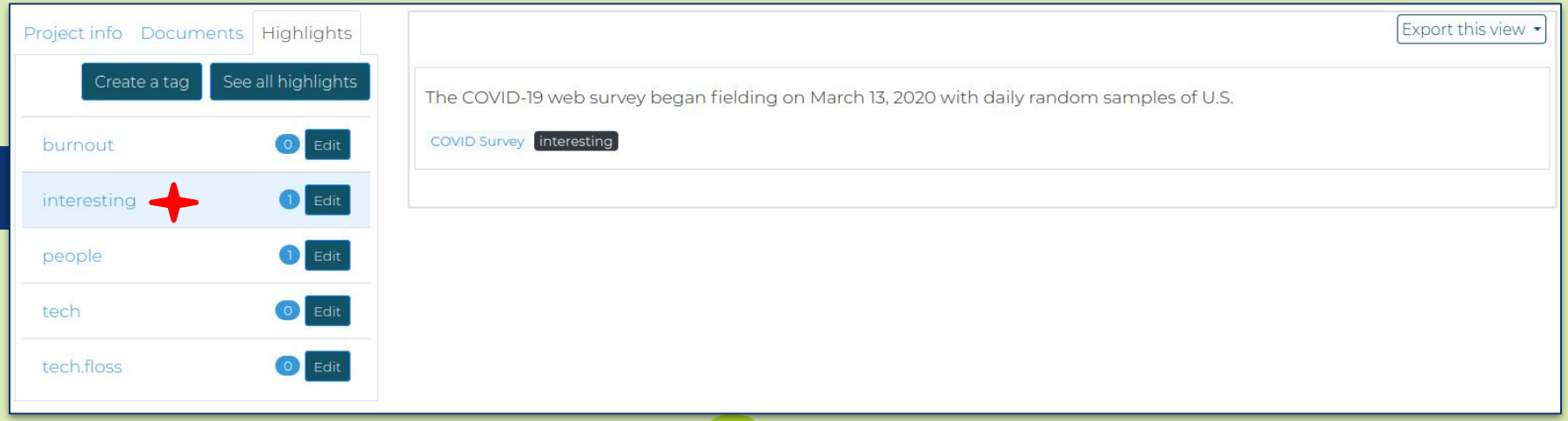
Seeing our work in the doc in TAGUETTE

Select Backlight if you want to grey out all non-highlighted text for a different view!



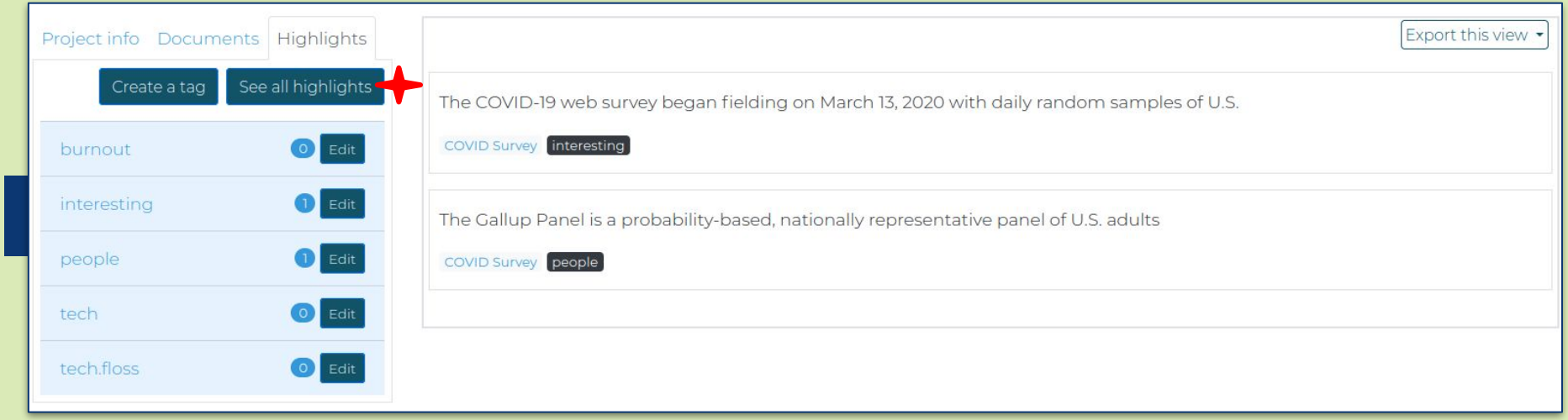
Seeing tagged highlights in TAGUETTE

- If you want to see all the highlighted text associated with a given tag, first go to the **Highlights** tab in the left pane.
- **Click a tag,** and in the right pane you will see a list of highlights with a blue link to the document where they originated from and all their associated tags with a black badge:



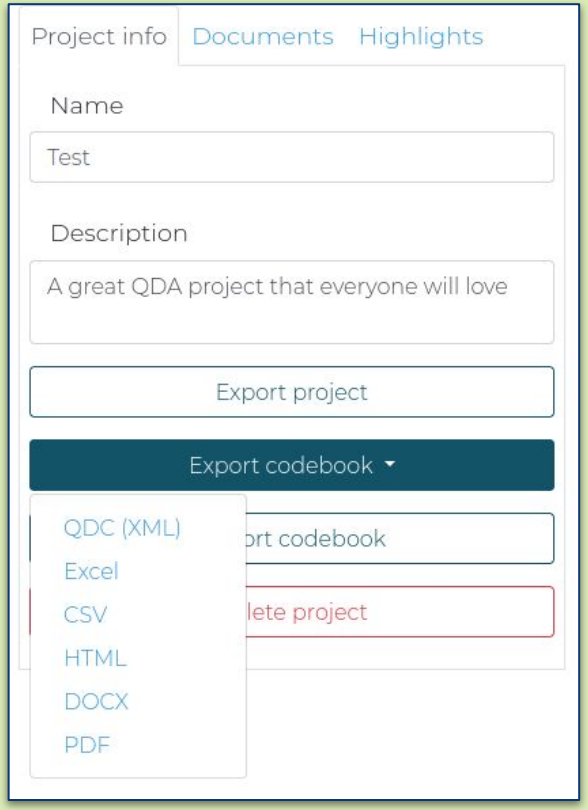
Seeing ALL of our highlights in TAGUETTE

If you want to see all the highlighted text for all documents and all tags, click **See all highlights** and you will see a list of quotes with a blue link to the document where they originated from and all their associated tags with a black badge:



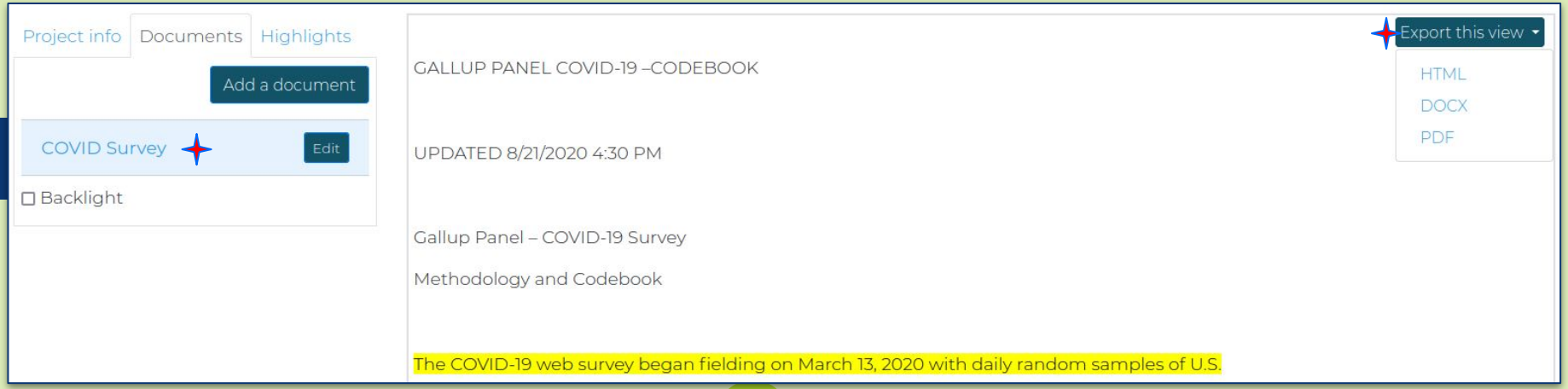
Exporting our codebook of TAGUETTE

- Go to the **Project** tab in the left pane of Taguette
- Click "**export codebook**" underneath the project description
- There is a variety of export options, including:
  - Excel & CSV (spreadsheet),
  - DOCX (Microsoft Word)
  - PDF
  - HTML,
  - QDC (CAQDAS standard codebook format)



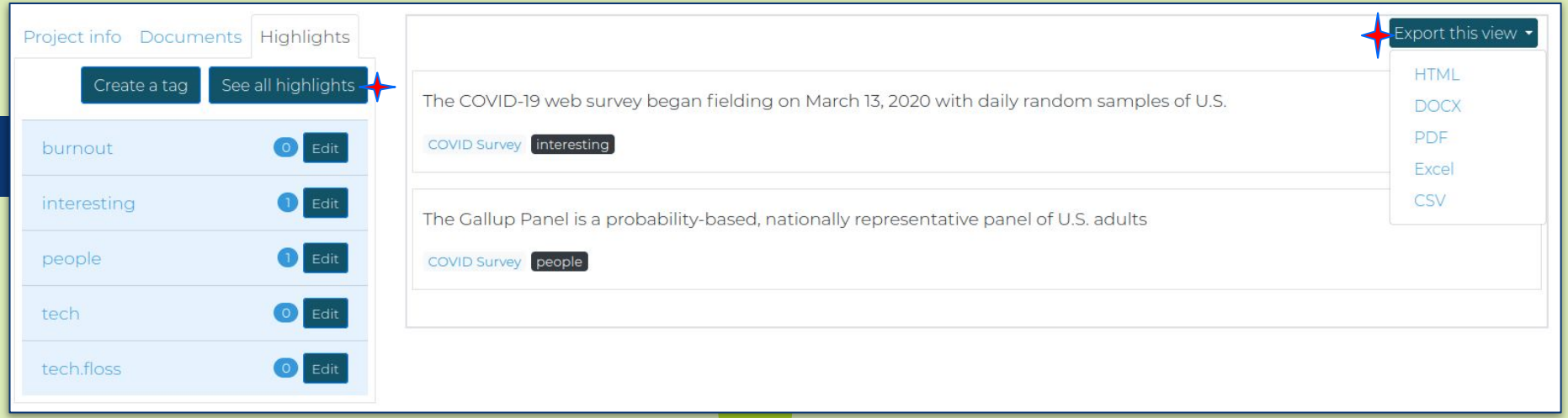
Exporting docs out of TAGUETTE

- Go to the **Documents** tab in the left pane, and click on the document
- Use the **Export this view** dropdown menu at the top right in the right pane.
  - Choice of HTML, DOCX, or PDF



Exporting ALL tags with highlights of TAGUETTE

- Go to the **Highlights** tab and click on **See all highlights**.
- Use the **Export this view** dropdown menu at the top right in the right pane.
  - DOCX, PDF, HTML, Excel, and CSV



What to do with CSV and Excel Exports

- ID and text
- Pivot tables / formulas to do analysis
  - guides.library.illinois.edu/qualitative/Taguette
- Adding columns of attributes
  - Adding back in your case classifications
- Adding weights
- Adding

	A	B	C	D
1	id	document	tag	content
2	264893	1	workflow	writes code and puts it o
3	264893	1	ghp	writes code and puts it o
4	264894	1	emotion-closed	we have GitHub as host
5	264894	1	emotion-open	we have GitHub as host
6	264894	1	resource-externa	we have GitHub as host
7	264894	1	ghp	we have GitHub as host
8	264894	1	resource-interna	we have GitHub as host
9	264894	1	workflow-collab	we have GitHub as host
10	264894	1	project-mgmt	we have GitHub as host
11	264894	1	communication	we have GitHub as host
12	264895	1	learning	it's more of like let's lear
13	264895	1	emotion-determi	it's more of like let's lear
14	264895	1	tools-workflow	it's more of like let's lear
15	264896	1	terminal	initially I would make a d
16	264896	1	workflow	initially I would make a d
17	264896	1	bestpractice	initially I would make a d
18	264897	1	workflow-collab	I was collaborating with

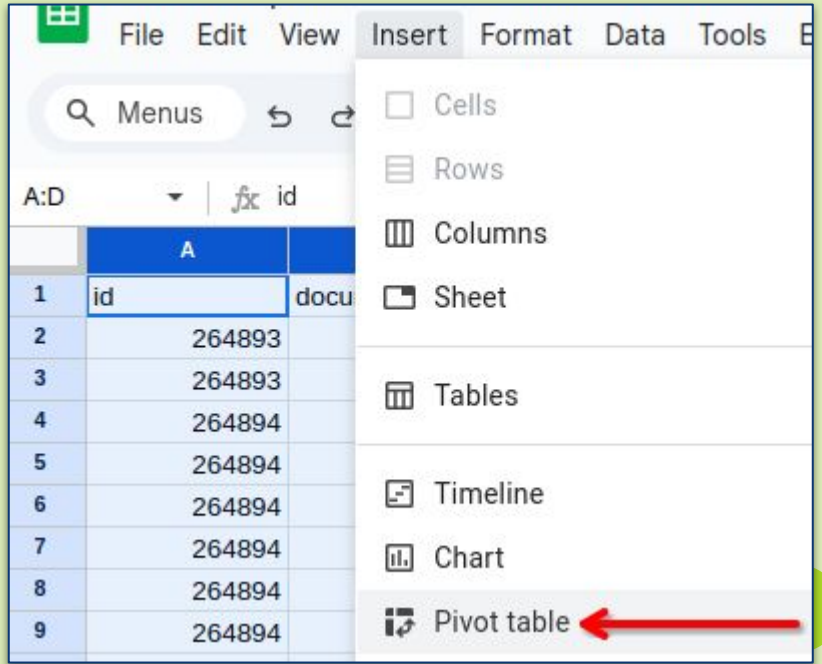
What to do with CSV and Excel Exports

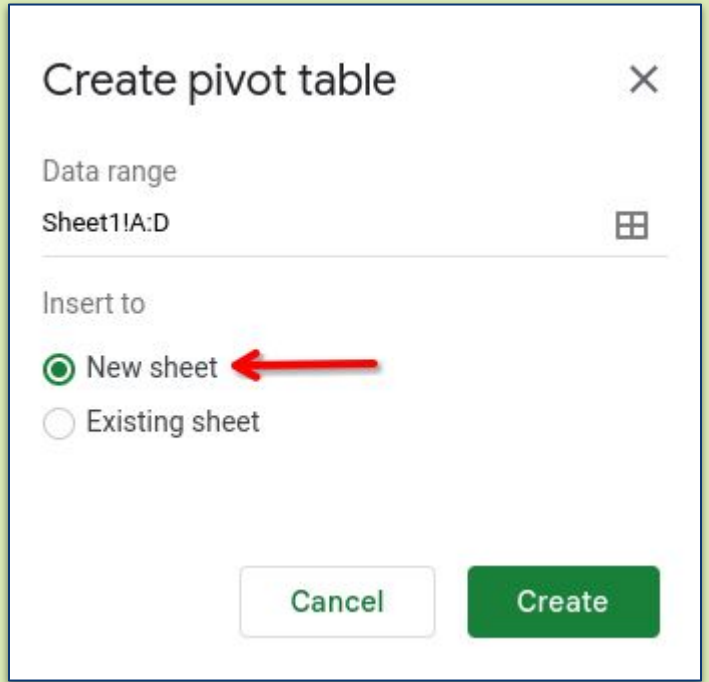
- **Create more columns** of interesting things you might want to query alongside the tags and highlighted content
  - If docs are done by-participant, it's simple to add case classifications as columns in an export
    - If docs have multiple individuals (e.g. focus group) it becomes complicated

	A	B	C	D	
1	id	document	inteviewee_status	tag	content
2	264893	ParticipantAlias00	Faculty	workflow	writes code c
3	264893	ParticipantAlias00	Faculty	ghp	writes code c
4	264894	ParticipantAlias00	Faculty	emotion-closed	we have GitH
5	264894	ParticipantAlias00	Faculty	emotion-open	we have GitH

Creating pivot tables to get counts from exports

- Selecting all your data from your export, go to Insert > Pivot Table
- Make sure you add it to a new sheet





Creating pivot tables to get counts from exports

COUNTA of tag	tag									
document	bestpractice	communication	conflict-burn-rep	emotion-anger	emotion-annoye	emotion-apathet	emotion-aware	emotion-certain	emotion-closed	emotion-confuse
1	9	3					9	4	3	5
2	5	5	1	3	3	2	7	8		7
3	3	2	1			6	1	5	3	1
4	6	5			2		4	7	1	1
6	12	12			2		19	8		
12	19	10			2	2	16	11		2
13	15	9		1	4	2	17	10	5	4
23	11	6					14	4	1	
30	18	5			2		25	8	2	1
37	12	3	1		3	2	24	5	1	1
40	13	7			1		21	3		2
Grand Total	**123**	**67**	**3**	**4**	**19**	**14**	**157**	**73**	**16**	**24**

Pivot table editor

Sheet1!A1:D2106

Clear all

Suggested

Rows Add

document

Order: Ascendi... Sort by: docume...

Show totals

Columns Add

tag

Order: Ascendi... Sort by: tag

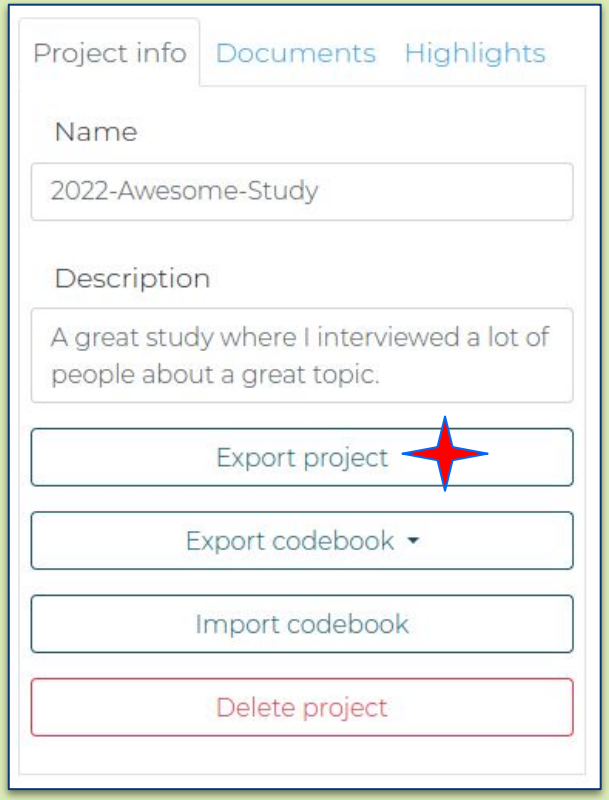
Show totals

Values Add

tag

Summarize by: COUNTA Show as: Default

Exporting projects



Taguette exports projects as SQLite files, which you can import into Taguette again or view in any software that supports it

The filename will be the date that you exported the file, followed by the name of the project:

2025-04-09_Test.sqlite

Query the SQLite exports!

SQLite can be used by a lot of different tools -- I like using DB Browser as a way to look at and query the data differently across the whole project file

DB Browser for SQLite - /home/vicky/Downloads/2025-04-09_Test.sqlite3

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: tags

Filter in any column

	id	project_id	path	description
	Filter	Filter	Filter	Filter
1	1	1	Parent	Further review required
2	4	1	Parent child	
3	5	1	Parent child grandchild	
4	2	1	Parent-child	
5	3	1	Parent-child-grandchild	
6	6	1	childhoodTrauma	description of childhood trauma

Sample SQL queries for a TAGUETTE export

Seeing the distribution of documents that are left-to-right or right-to-left across a project:

```
SELECT text_direction, COUNT(text_direction) AS count
FROM documents GROUP BY text_direction
```

Sample SQL queries for a TAGUETTE export

Seeing what time of day had the most project activity, by hour:

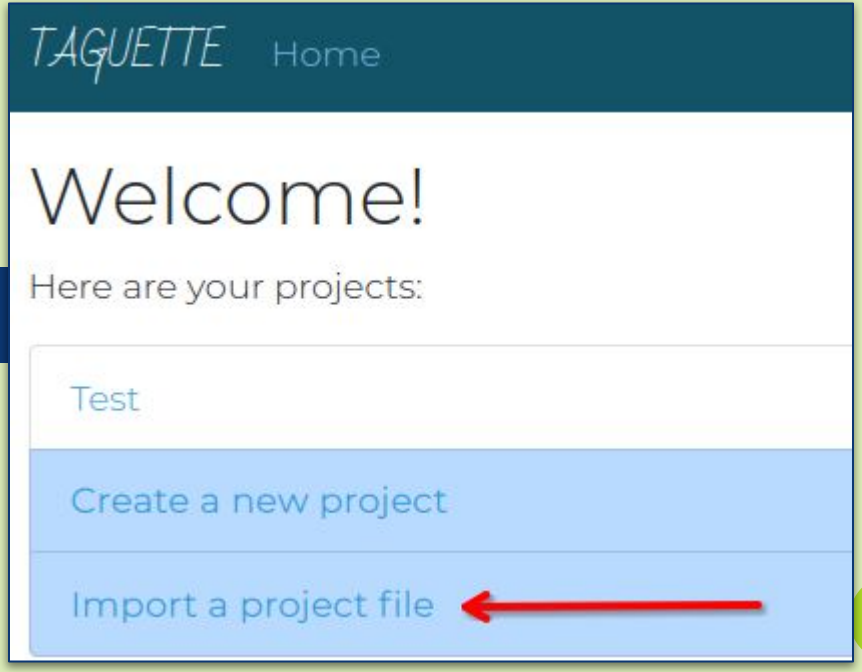
```
SELECT strftime ('%H',date) AS hour, COUNT(*) AS actionsCount
FROM commands
GROUP BY hour
ORDER BY hour ASC;
```

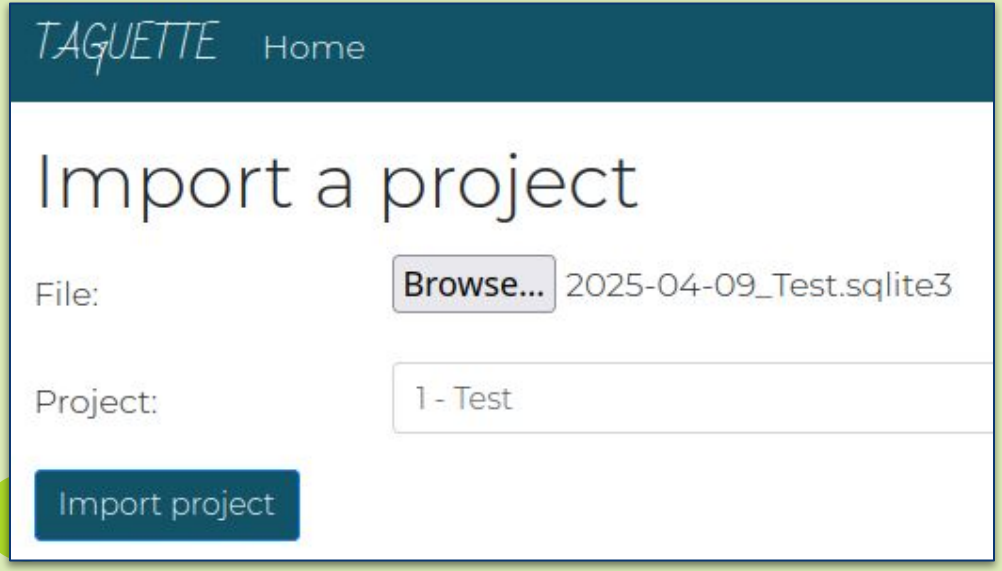
Seeing the timeline of actions across the whole project:

```
SELECT DATE(date) AS date_only, COUNT(*) AS actionsCount
FROM commands
GROUP BY DATE(date);
```

Importing projects from SQLite file

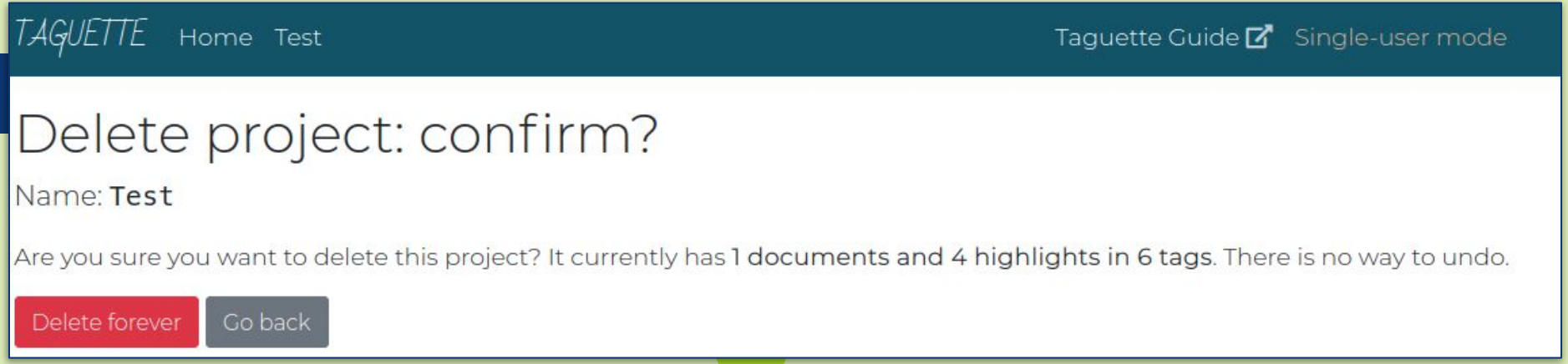
You can import an exported Taguette project back into a Taguette instance, either the same one or a completely different one! Useful for going from the Web to local computer or vice versa





Deleting a project is permanent

- In the Project tab of the left pane, select **Delete a project**
- Taguette will show you a warning before you do so, in case you need to export any thing
- **Deletions are permanent** and there is no way to undo this



Summary & Q+A

Summary

Taguette is a straight up QDA tool

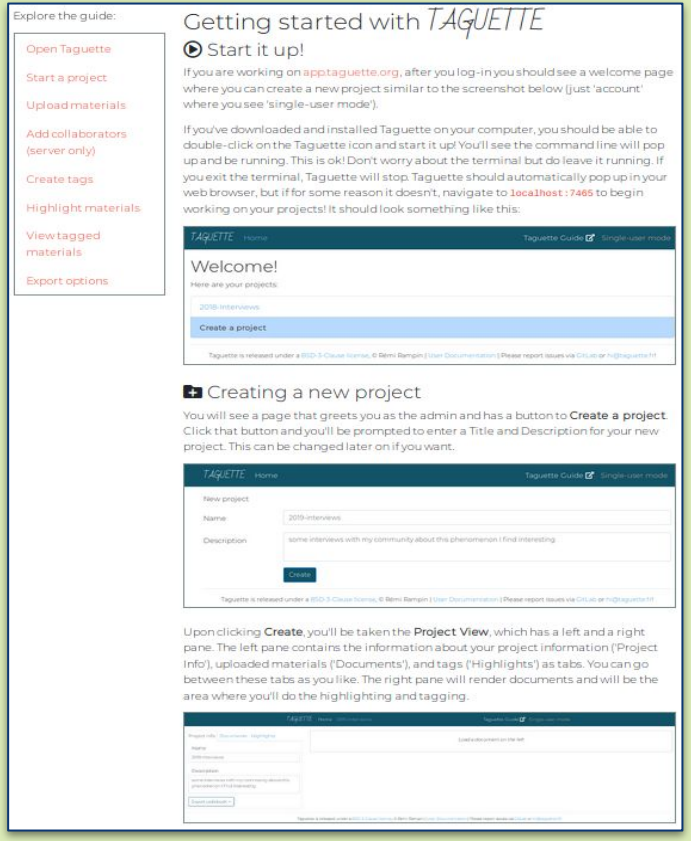
Very simple by design; meant for highlighting text and light querying; any more analyses are facilitated through the various exports

Taguette is one of few free and open source CAQDAS

It’s important to foster free and open source tooling for qualitative analysis, and we’re trying to do that with this project

Community outreach & support for TAGUETTE

- Inbox: hi@taguette.org
- Getting Started guide: taguette.org/getting-started.html
- Presentations on the OSF: osf.io/xdszm
- Listservs for announcements: groups.io/g/taguette-announce
- Listserv to connect users together: groups.io/g/taguette



Thank you!
I'd love to hear questions!

Vicky Rampin (she | they)

hi@taguette.org

Try out Taguette – taguette.org